Pattern Recognition and Machine Learning (EE602L)

Lab 4: Multiclass Logistic Regression and Support Vector Machines

Instructions:

- Use Google Colab.
- Submit one well-commented notebook (Lab4-RollNum.ipynb) along with a PDF report (Lab4-RollNum.pdf) containing experimental outputs, analysis of the results, and conclusions.
- Wrong file names or missing files won't be graded.
- **Deadline:** 07/09/2026 11.59 PM. Late submissions lose 10% per day.
- This is an individual assignment. Please follow the Institute's Academic Code of Honor.

Question 1 — Multiclass Logistic Regression via Softmax

Problem Statement

Dataset: Wine dataset, available directly through scikit-learn (from sklearn.datasets import load_wine) — 13 numerical features, 3 classes.

Implement multiclass logistic regression using the softmax function from scratch, and train it on the Wine dataset using batch gradient descent. Evaluate the model on held-out data and compare its performance with a library implementation. Verify numerically that the softmax formulation reduces to the binary sigmoid case when $K = 2$.

(a) Data preparation

Load the Wine dataset and inspect the class balance and summary statistics. Split the data into training, validation, and test sets in a 70:15:15 ratio, implementing the split from scratch (NumPy/Pandas only) using a fixed random seed of 42 and stratified sampling. Standardize the features using training-set statistics only, and one-hot encode the target for training while retaining integer labels for evaluation.

(b) Model implementation

Implement the softmax function, the multiclass cross-entropy cost, and its gradient from scratch, and train the model using batch gradient descent with at least three learning rates. Select the best learning rate using validation cross-entropy. Verify numerically (e.g. on a 2-class subset, or by fixing $\theta_2 = 0$) that softmax reduces to the binary sigmoid when $K = 2$, and show this check in your notebook.

(c) Evaluation

Predict each test sample's class as $\arg\max_k P(y = k \mid x)$, and report a confusion matrix, accuracy, macro-averaged precision/recall/F1, and multiclass log-loss computed manually. Compare your from-scratch model against sklearn.linear_model.LogisticRegression() on the test set, and discuss any differences.

Question 2 — Support Vector Machines

Problem Statement

Implement and evaluate linear Support Vector Machine (SVM) classifiers using synthetic and real-world datasets. Investigate the geometric interpretation of margins and support vectors, and study how the regularization parameter C affects the decision boundary and classification performance.

(a) Geometric intuition on synthetic 2D data

Generate a small 2D synthetic dataset (e.g. make_blobs or make_classification) using a fixed random seed of 42, fit a linear SVM using SVC(kernel='linear'), mark the support vectors, and plot the decision boundary together with both margin boundaries. Introduce class overlap and repeat using at least three values of C, then compare the resulting boundaries and discuss the effect of C on margin width and misclassification.

(b) Effect of regularization on classification

Generate a new 2D synthetic dataset with overlapping classes (e.g. make_classification with increased class overlap) using a fixed random seed of 42, and split the dataset into training and validation sets in an 80:20 ratio using the same seed. Train linear SVM classifiers on the training set for at least five values of C spanning several orders of magnitude. For each value, visualize the resulting decision boundary and margin, and identify the support vectors. Compare training and validation performance across the different values of C, and discuss how increasing or decreasing C changes the trade-off between margin width and classification error.

(c) Full evaluation on a real dataset

Load the Breast Cancer Wisconsin dataset (sklearn.datasets.load_breast_cancer), split it 70:15:15 using a fixed random seed of 42 and stratified sampling, and standardize using training-set statistics. Train a linear SVM and select C using validation performance, then report the selected value of C and the number of support vectors for the selected model. On the test set, evaluate the selected linear SVM, reporting accuracy, precision, recall, F1-score, and ROC-AUC (computed using the decision function scores), and compare performance across the different values of C considered; discuss the effect of C and regularization on the decision boundary and generalization.